Radio button

Radio buttons let people select one option from a set of options

Resources +2

Use cases

People should be able to do the following with assistive technology:

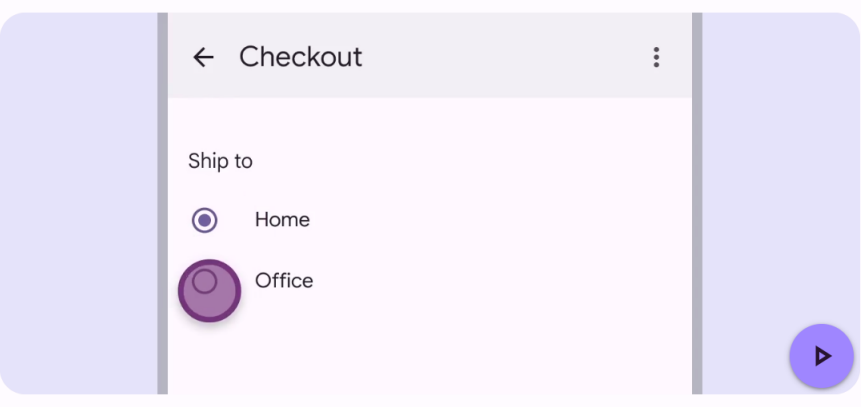
- Navigate to a radio button
- Select a radio button
- Get appropriate feedback based on input type

Interaction & style

A radio button can be either selected or unselected. Selecting one radio button deselects any others. A radio group can start with one radio button selected, or none selected.

Once a radio button is selected, the group can't be deselected. To let people opt out of their selection, either provide a **Not applicable** or **No option** radio button, or provide a separate way to deselect all radio buttons, like **Clear selection**.

People should be able to select either the text label or the radio button to select an option.



Only one radio button is selected at a time

Avoid applying density by default

Don't apply density to radio buttons by default. This lowers their targets below Material's recommendation of 48x48 CSS pixels. Instead, give people a way to choose a higher density, like selecting a denser layout or changing the theme.

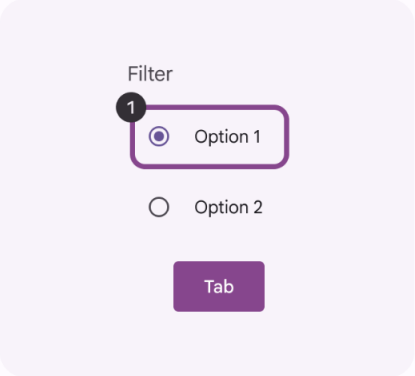
To ensure this density setting can be easily reverted when it's active, keep all targets to change it at a minimum of 48x48 CSS pixels each.

Initial focus

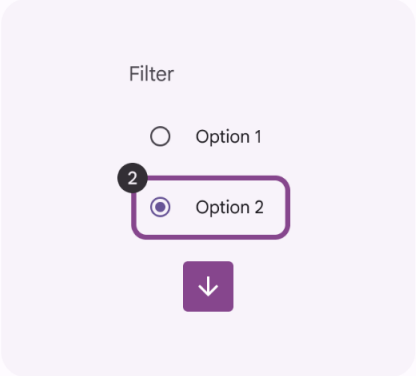
When outside the radio group, **Tab** moves focus directly to the selected radio button, or the first one if none are selected.

Shift+Tab instead focuses on the last radio if none are selected.

Use the **arrows** to navigate between options.



Tab brings the focus to the initially selected item or the initial radio option



Arrows move to next element in a list

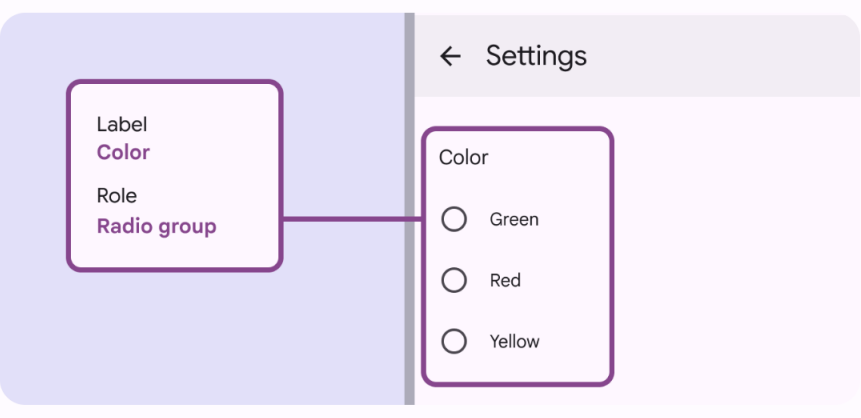
Keyboard navigation

Keys	Actions
Tab	Moves focus into the group to the selected radio button, or the first if none are selected
Shift + Tab	Moves focus into the group to the selected radio button, or the last if none are selected
Arrows	Moves focus and selects the previous or next radio button. Wraps focus and selection between the first and last radio buttons.
Space	Selects a focused radio button. If already selected, does nothing.

Labeling elements

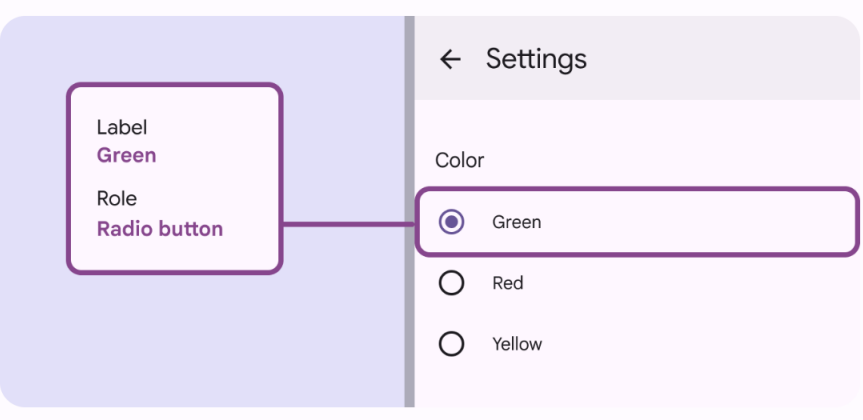
If the UI text is correctly linked to the radio button, assistive tech such as a screenreader will read the UI text, followed by the component's role.

The accessibility label for a group of radio buttons is typically the same as its title. The role is **Radio group**.



Label the radio group based on the category title

The accessibility label for an individual radio button is typically the same as its adjacent text label.



Label the radio button based on its label text